

PROBLEM SOLVING AND TESTING USING JAVA

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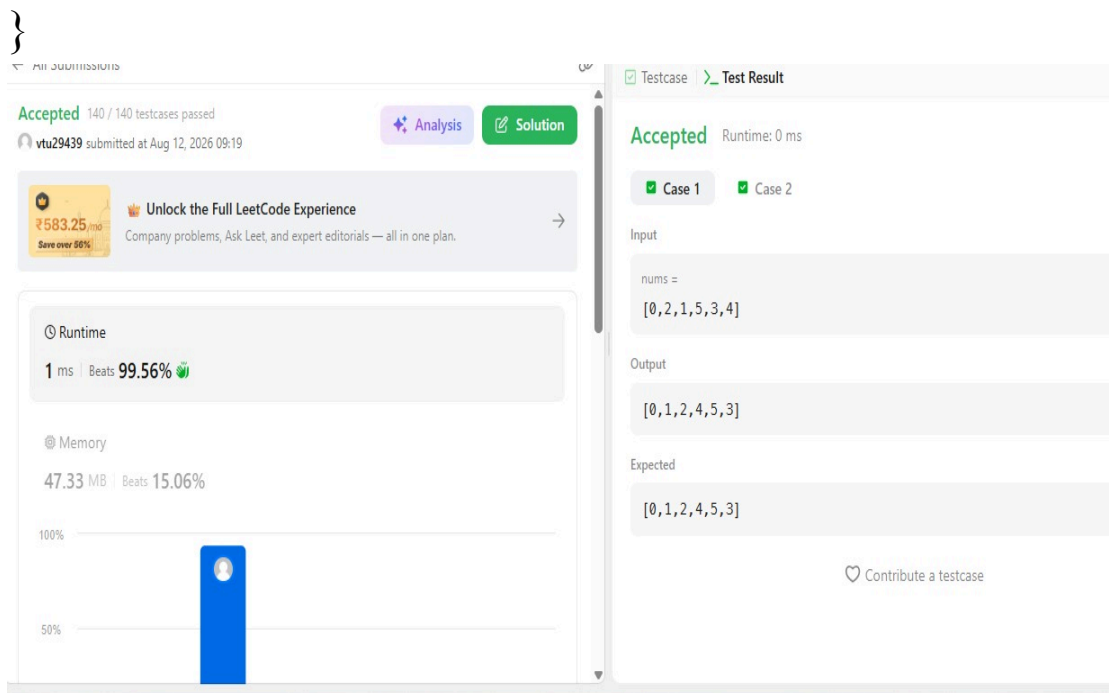
VTUNO: 28767

WEEK-2 PROBLEMS

1. Build Array from Permutation- <https://leetcode.com/problems/build-array-from-permutation/>

CODE:

```
class Solution {  
    public int[] buildArray(int[] nums) {  
        int n = nums.length;  
        int[] ans = new int[n];  
  
        for (int i = 0; i < n; i++) {  
            ans[i] = nums[nums[i]];  
        }  
  
        return ans;  
    }  
}
```



2. Shuffle the Array – <https://leetcode.com/problems/shuffle-the-array/>

CODE:

```
class Solution {
    public int[] shuffle(int[] nums, int n) {
        int[] result = new int[2 * n];

        for (int i = 0; i < n; i++) {
            result[2 * i] = nums[i];    // xi
            result[2 * i + 1] = nums[i + n]; // yi
        }
    }
}
```

```

        return result;
    }
}

```

All Submissions

Pending... Speed Up

vtu29439 submitted at 2 minutes ago

* preparing runtime environment

Code | Java

```

1 class Solution {
2     public int[] shuffle(int[] nums, int n) {
3         int[] result = new int[2 * n];
4
5         for (int i = 0; i < n; i++) {
6             result[2 * i] = nums[i]; // xi
7             result[2 * i + 1] = nums[i + n]; // yi
8         }
9
10        return result;
11    }
12}

```

View more

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

nums =
[2, 5, 1, 3, 4, 7]

n =
3

Output

[2, 3, 5, 4, 1, 7]

Expected

[2, 3, 5, 4, 1, 7]

3. Find the Highest Altitude – [https://leetcode.com/problems/find-the-higher-altitude/](https://leetcode.com/problems/find-the-highest-altitude/)

CODE:

```

class Solution {
    public int largestAltitude(int[] gain) {
        int currentAltitude = 0;
        int maxAltitude = 0;

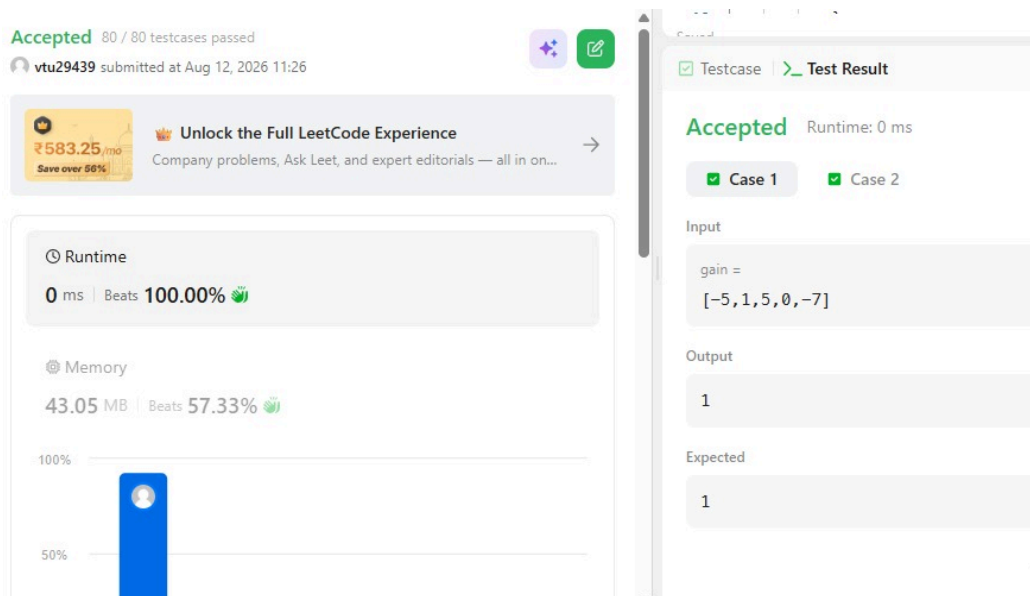
```

```

    for (int g : gain) {
        currentAltitude += g;
        if (currentAltitude > maxAltitude) {
            maxAltitude = currentAltitude;
        }
    }

    return maxAltitude;
}
}

```



Accepted 80 / 80 testcases passed
vtu29439 submitted at Aug 12, 2026 11:26

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Runtime
0 ms | Beats 100.00%

Memory
43.05 MB | Beats 57.33%

Testcase Test Result
Accepted Runtime: 0 ms
Case 1 Case 2

Input
gain =
[-5, 1, 5, 0, -7]

Output
1

Expected
1

4. Java Dequeue – <https://www.hackerrank.com/challenges/java-dequeue/problem>

CODE:

```
import java.util.*;

public class test {

    public static void main(String[] args) {

        Scanner in = new Scanner(System.in);

        if (!in.hasNextInt()) return;

        int n = in.nextInt();

        int m = in.nextInt();

        Deque<Integer> deque = new ArrayDeque<>();

        Map<Integer, Integer> countMap = new
HashMap<>();

        int currentUnique = 0;

        int maxUnique = 0;

        for (int i = 0; i < n; i++) {

            int num = in.nextInt();

            // Add the new element to the sliding window
            deque.addLast(num);
```

```
countMap.put(num,  
countMap.getOrDefault(num, 0) + 1);
```

```
if (countMap.get(num) == 1) {  
    currentUnique++;  
}
```

// If the window exceeds size m, remove the
oldest element from the front

```
if (deque.size() > m) {  
    int removedNum = deque.removeFirst();  
    countMap.put(removedNum,  
countMap.get(removedNum) - 1);
```

```
    if (countMap.get(removedNum) == 0) {  
        currentUnique--;  
        countMap.remove(removedNum);  
    }  
}
```

// Once the window reaches size m, check and
update maxUnique

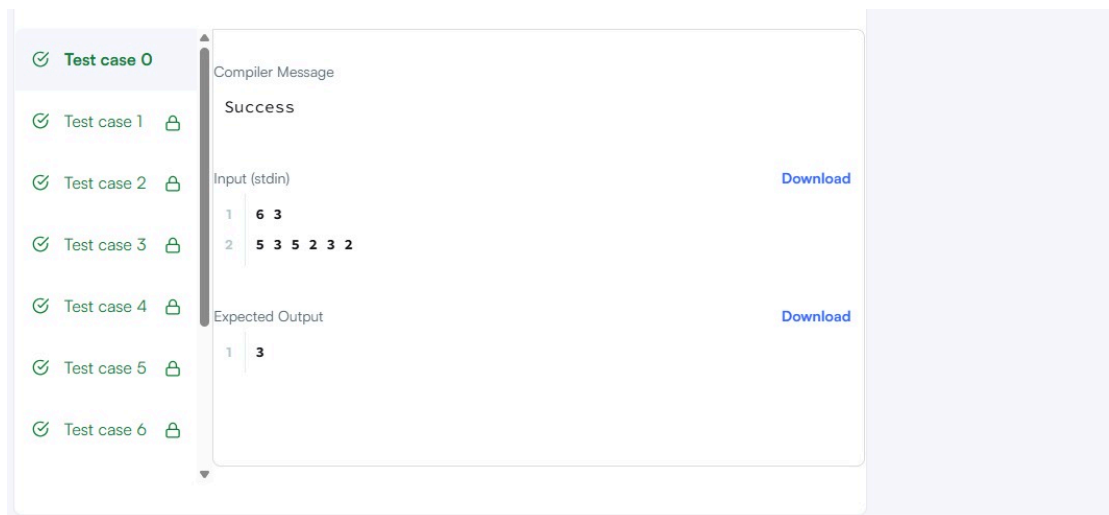
```
if (deque.size() == m) {
```

```

        if (currentUnique > maxUnique) {
            maxUnique = currentUnique;
        }
    }
}

System.out.println(maxUnique);
in.close();
}
}

```



5. Java HashSet – <https://www.hackerrank.com/challenges/java-hashset/problem>

CODE:

```
import java.io.*;
import java.util.*;
import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        int t = s.nextInt();
        String [] pair_left = new String[t];
        String [] pair_right = new String[t];

        for (int i = 0; i < t; i++) {
            pair_left[i] = s.next();
            pair_right[i] = s.next();
        }

        HashSet<String> pairs = new HashSet<String>();

        for (int i = 0; i < t; i++) {
            pairs.add(pair_left[i] + " " + pair_right[i]);
        }
    }
}
```

```
        System.out.println(pairs.size());
    }
}
}
```

Congratulations!

You have passed the sample test cases. Click the submit button to run your code against all the test cases.

✓ Sample Test case 0

Your Output (stdout)

1	1
2	2
3	2
4	3
5	3

Expected Output

1	1
2	2
3	2
-	-

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THANK YOU